



ECOSYSTEM

A collection of living organisms that share a habitat and are reliant on each other for survival.

ELECTRICITY

Anything related to electric charge. Electric current is the movement of particles with electric charge. Electricity is used to power homes, cars, and many other modern machines.

ELECTROLYSIS

A process that divides substances into simpler parts using electric current.

ELECTROMAGNETIC SPECTRUM

The whole range of different types of electromagnetic radiation, from gamma rays to radio waves. Visible light, which we can see, is part of the electromagnetic spectrum.

ELECTRON

A negatively charged particle found in the outer part of an atom. Moving electrons carry electricity and cause magnetism.

ELEMENT

A substance made of only one kind of atom. There are 118 known elements, about 90 of which occur naturally.

ENERGY

What enables work to be done. Energy exists in many different forms, such as electrical and chemical. It cannot be created or destroyed—only transferred.

EVAPORATION

The process by which atoms or molecules of a liquid turn into a gas.



EVOLUTION

The process by which species gradually change and adapt to a changing environment over very long periods of time.

FAT

An energy-rich substance found in living things. In the body, fats are used to store energy, surround nerves, and as a layer of insulation.

FORCE

A push or a pull that causes an object to change its speed, direction, or shape.

FOSSIL FUEL

A fuel derived from the fossilized remains of living things. Fossil fuels include coal, oil, and gas. They release carbon dioxide and other harmful gases when they are burned, contributing to global warming.

FULCRUM

The pivot or fixed point around which a lever turns.

FUNGUS

A group of microorganisms that feed on plants and animals. They break down dead animals and plants, feeding on the nutrients.

GAS

A state of matter where the particles are spaced out and move around at high speed. Gases flow to fill a container and can be compressed.

GENE

A length of DNA that carries a code to perform a specific job. Genes instruct cells to make proteins that affect one or more of the organism's characteristics. Genes are passed down from one generation to the next.



GENERATOR

A device that produces electrical energy from kinetic energy. Wind turbines and many power plants used generators.

GLOBAL WARMING

A rise in the average temperature of Earth's atmosphere, caused by rising levels of carbon dioxide and other greenhouse gases. Global warming has many effects on Earth's climate, including the melting of ice at the poles, a rise in sea levels, and the occurrence of more extreme weather events.

GRAVITY

A force that pulls all things with mass toward each other. On Earth, gravity pulls objects to the ground and gives them weight. The planets of the Solar System are kept in orbit by gravity.

GREENHOUSE GAS

A gas in the atmosphere that traps the Sun's warmth around the planet. Greenhouse gases include carbon dioxide and methane.

HABITAT

The natural home of an animal or plant. It can be as small as an underside of a leaf and as vast as an entire forest.

IMMUNE SYSTEM

The defense mechanism of the human body that protects us from diseases. Every time you sneeze, cough, or get a cold or fever, it is your immune system fighting off a threat.

INFRASTRUCTURE

The basic facilities, such as buildings, roads, and bridges, required by a country or society to work.



INSULATOR

A material through which heat or electricity cannot pass easily.

ISOTOPES

Different versions of an element, each with the same number of protons but different numbers of neutrons in their nuclei.

LASER

(Light amplification by stimulated emission of radiation.) A device that creates a beam of light with waves that are in step and of a precise wavelength.

LENS

A glass or plastic object with curved surfaces that cause light to bend as it passes through. A lens that curves inward is called concave, and a lens that curves outward is called convex. Lenses are used to form images in cameras, telescopes, and microscopes.

LIQUID

A state of matter where the atoms or molecules are closely packed together. The bonds between them are stronger than in gases but weaker than in solids, allowing the particles to move freely. Liquids flow and take the shape of a container and cannot be compressed.

MAGNET

An object that has a magnetic field and attracts or repels other magnetic objects. Things are attracted or repelled by magnets due to an invisible force called magnetism.

MAGNETIC FIELD

The invisible pattern of force that stretches out around a magnet.